

EDUGENIE: GOOGLE GEMINI POWERED LEARNING ASSISTANT

PROJECT DOCUMENTATION REPORT

Project Status: COMPLETED

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1. Project Description

EduGenie is a lightweight AI-powered educational assistant that simplifies learning through generative AI. It enables students to ask questions and receive smart answers, understand complex concepts through simplified explanations, generate quizzes, receive personalized learning recommendations, and summarize educational passages.

2. Objectives

- Provide simple and concise answers to academic questions.
- Explain difficult concepts in beginner-friendly language.
- Generate multiple-choice quizzes from educational text.
- Summarize lengthy educational passages for quick revision.
- Provide structured learning recommendations from beginner to advanced levels.

3. Technologies and Prerequisites

- Python 3.10+
- FastAPI Framework
- HTML5 and CSS3 Interface
- Google Gemini API
- Uvicorn ASGI Server
- Jinja2 Templating Engine
- Google AI Studio Key

4. System Architecture

The project uses a FastAPI backend with a simple HTML/CSS frontend. The backend connects the user interface to the AI-powered modules. Gemini is used for question answering, summarization, quiz generation, and learning-path recommendations, while the project document specifies LaMini-Flan-T5 for concept explanation.

5. Folder Architecture

```
EduGenie/  
■■■ main.py  
■■■ explanation_module.py  
■■■ qna.py  
■■■ quiz_module.py  
■■■ summary_module.py  
■■■ learning_path.py  
■■■ templates/  
■   ■■■ index.html  
■■■ static/  
■   ■■■ style.css  
■■■ requirements.txt
```

6. Core Functionalities

Explanation Module: Provides simplified explanations of complex concepts.

Q&A; Module: Answers general knowledge and academic questions using Gemini.

Quiz Module: Generates three MCQs with four options from a supplied passage and returns structured output.

Summary Module: Converts long educational passages into concise, easy-to-revise content.

Learning Path Module: Generates a structured beginner-to-advanced learning path with useful resources and guidance.

7. Backend API

The FastAPI backend defines RESTful endpoints to process tasks cleanly:

Endpoint	Method	Description
/qa	POST	Handles academic Q&A queries
/explain	POST	Provides concept explanations
/quiz	POST	Generates 3 MCQs from text
/summarize	POST	Summarizes text passages
/learn/recommendations	POST	Creates learning roadmaps

8. Frontend Interface

The web interface contains a task selection dropdown, a text input area, a submit button, and a result area. The form sends asynchronous requests to the FastAPI backend and displays the generated result cleanly below the input area.

9. Running the Project

1. Install Python 3.10 or newer.
2. Install required packages using requirements.txt.
3. Configure the Gemini API key in the .env file.
4. Start server: `uvicorn main:app --reload`
5. Open browser at: `http://127.0.0.1:8000`

10. Testing

Test the application by asking an academic question, requesting an explanation of a topic, generating a quiz from a passage, summarizing a long paragraph, and requesting a personalized learning plan.